OMFS 711

Principals of Oral Surgery

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Armamentarium for Basic Oral Surgery

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Learning Objectives & Competency Statements

Learning objectives:

- List commonly utilized instruments in basic oral surgical procedures
- Describe different parts of the instruments and instrument handling
- Understand the design of instruments and how they adapt to tooth anatomy

Required Reading

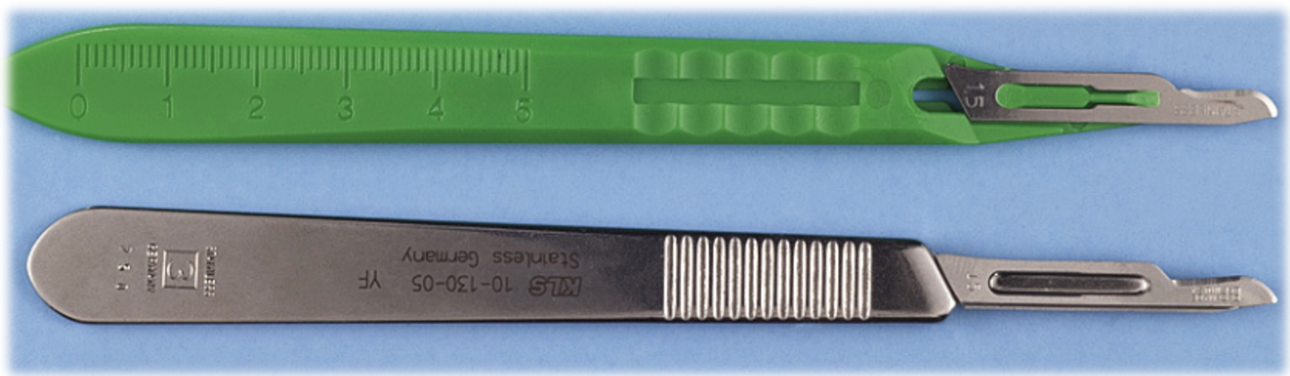
Chapter 7

Instrumentation for Basic Oral Surgery

Hupp, James, R. et al. Contemporary Oral and Maxillofacial Surgery. 7th Edition

Instruments to Incise Tissue

- ***Scalpel***
  - Primary instrument for making incisions
  - Is composed of
    - A handle
    - A sharp sterile blade
  - Single use (plastic handle)
  - Reusable handle
    - No. 3 handle
      - most common in oral surgery

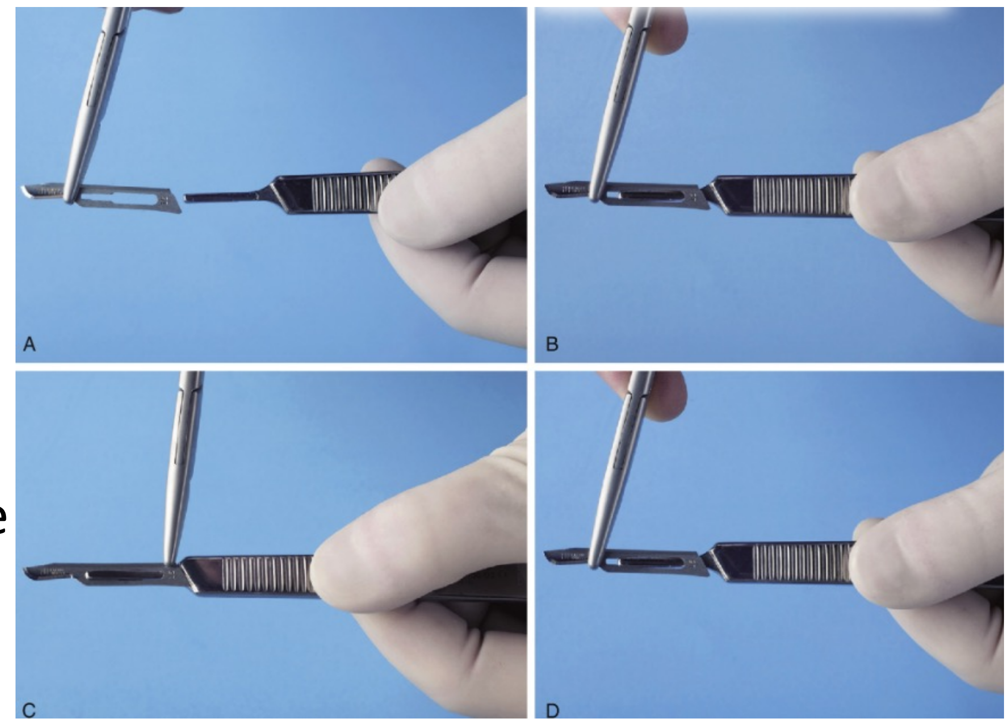


Instruments to Incise Tissue

- ***Scalpel blades***
  - #15 -most often used blade in intraoral surgeries
    - blade is small and is used to make sulcular incisions and through soft tissue
  - #12 is used for
    - Mucogingival procedures
    - Incision on distal of teeth or in maxillary tuberosity area
  - #11 has sharp-pointed tip
    - commonly used for small stab incisions (i.e., incising abscess to allow drainage)

Instruments to Incise Tissue

- ***Inserting/removing scalpel blades***
  - blade is held along the unsharpened edge, where it is reinforced with a small rib
  - male portion of the handle is kept upward
  - blade is then slowly slid onto the handle along the grooves in the male portion until it clicks into position



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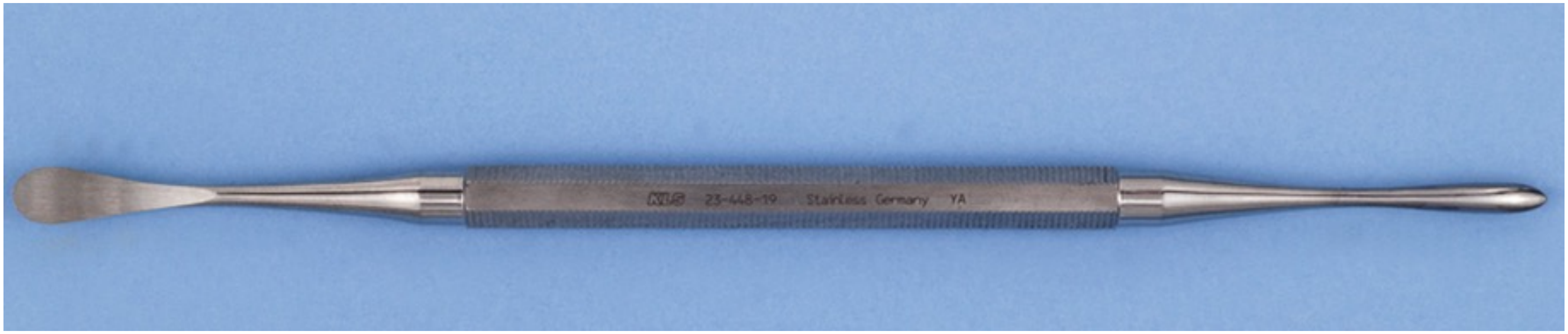
Instruments to Incise Tissue

- ***<u>Holding Scalpel</u>***
- Handle is held in pen grasp
- Mobile tissue should be held firmly in place under some tension to maximize cutting efficiency
- For full-thickness mucoperiosteal incision, blade should be pressed down firmly so that the mucosa and periosteum is incised with a single stroke

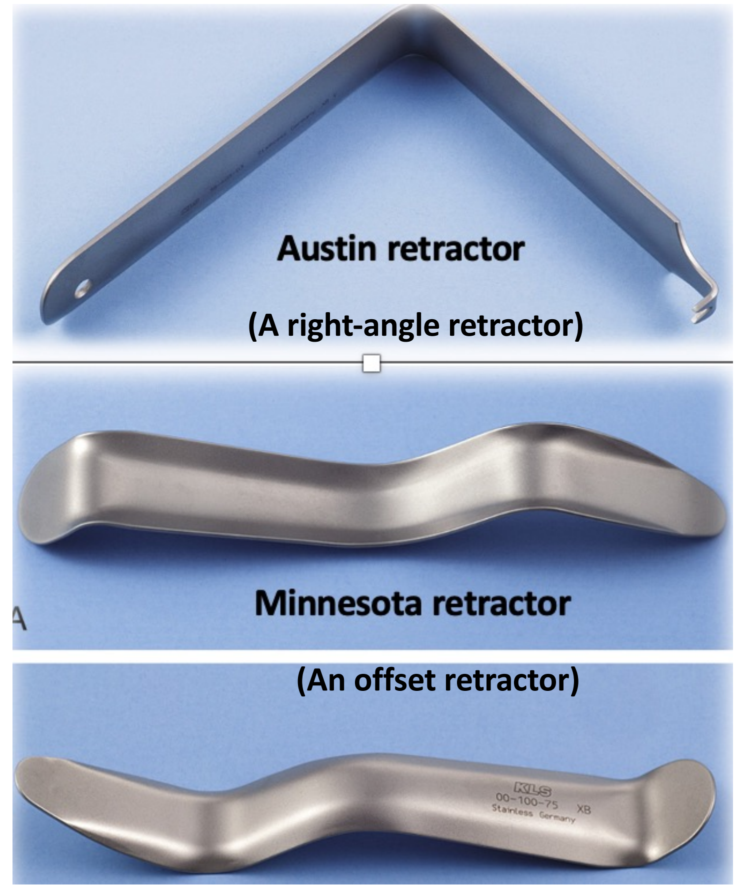
Instruments to Elevate Mucoperiosteum



- *No. 9 Molt periosteal elevator** is the most common instrument
- Pointed end is used to reflect periosteum from teeth, starting at *dental papillae*
- Broad, rounded end is used to elevate periosteum from bone

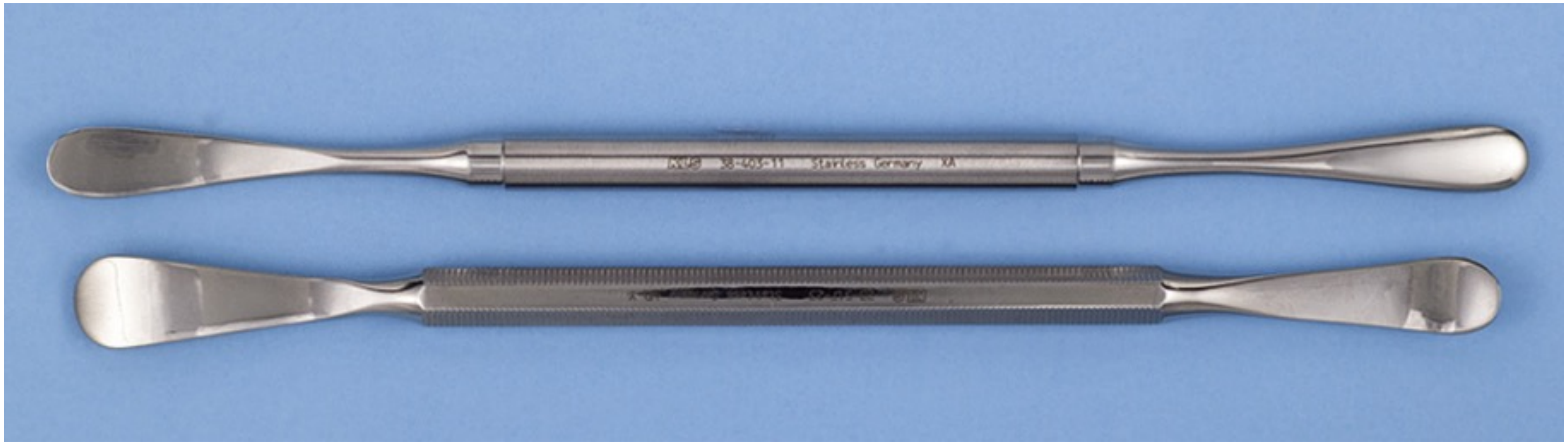
Instruments to Retract Soft Tissue

- ***<u>Retractors</u>***
  - Provide access and visibility during surgery, and act as shield
  - Specifically designed to
    - Retract cheek
    - Retract Tongue
    - Retract mucoperiosteal flap
    - Protect the soft tissue from sharp cutting or rotary instruments



Instruments to Retract Soft Tissue

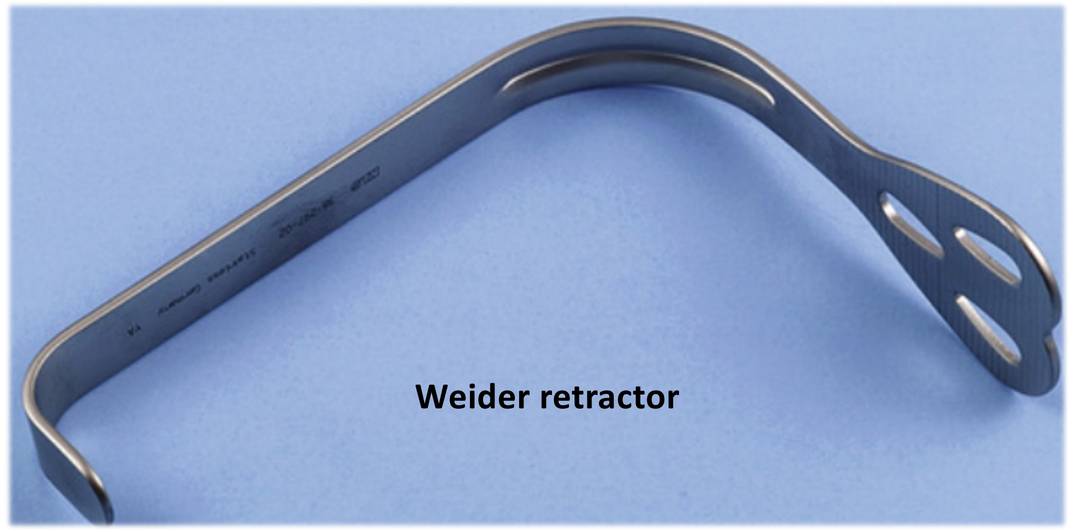
- ***Retractors***



The Henahan (top) and Seldin (bottom) retractors are broader instruments that provide broader retraction and increased visualization

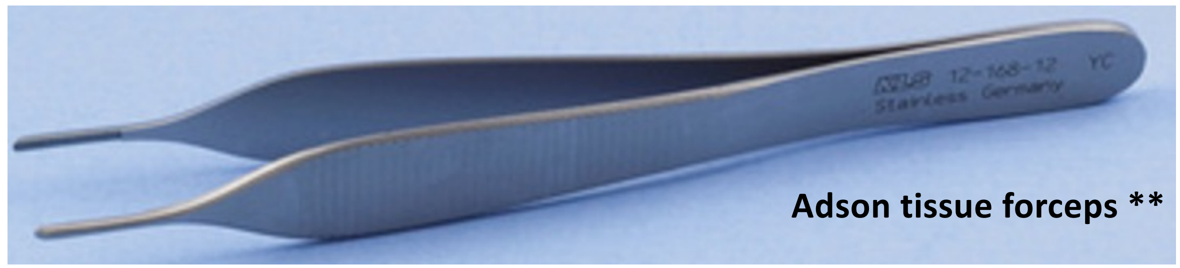
Instruments to Retract Soft Tissue

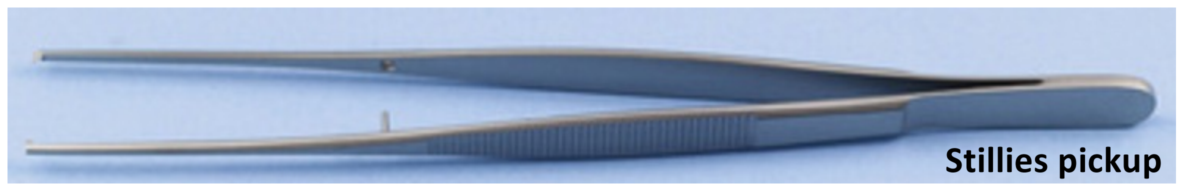
- ***Retractors***
  - Weider retractor is a large retractor designed to retract the tongue away from the surgical field
  - The serrated surface helps engage the tongue so that it can be held securely



Instruments to Grasp Soft Tissue

- ***<u>Soft Tissue Forceps</u>***
  - Useful to grab soft tissue during
    - Incision and flap elevation
    - Suturing
  - Locking type
    - Helpful to stop bleeding



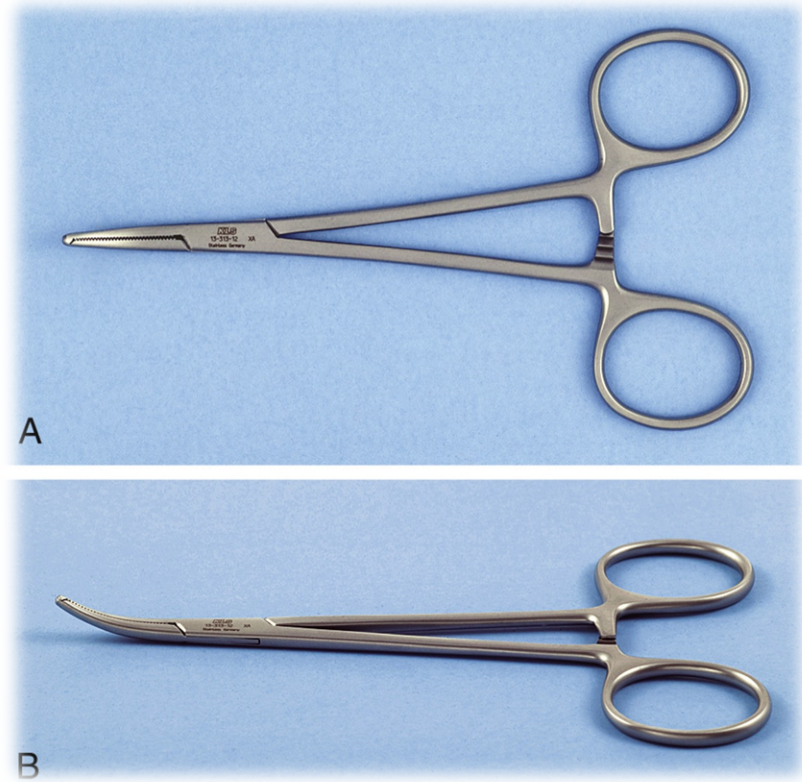


Instruments to Grasp Soft Tissue

- ***<u>Allis Tissue Forceps</u>***
  - locking forceps are necessary during removing larger amounts of tissue or doing biopsies
  - locking handle can be held by assistant to apply necessary tension for proper dissection of the tissue

Instruments to Control Hemorrhage

- ***<u>Hemostats</u>***
  - used when pressure alone does not stop the bleeding from a large artery or vein
  - come in a variety of shapes
    - Small or large
    - Straight or curved



Instruments to Remove Bone

- ***Rongeurs****
  - Bone is cut when sharp blades are squeezed together
  - Two major designs for rongeur forceps
    - Side-cutting forceps
    - Side- and end-cutting forcep

Instruments to Remove Bone

- *__Burr and Handpiece__*
  - Used to remove bone in surgical extraction
  - No. 557 or No. 703 fissure burr and No. 8 round burr are commonly used
  - For large amount of bone removal, a large-bone burr that resembles an acrylic burr is typically used

557

703

Instruments to Remove Bone

- ***Mallet and Chisel***
    - Used for removing lingual tori
    - Sectioning teeth

Instruments to Remove Bone

- ***Bone File***
  - Double ended instrument
    - Large end + small end
    - Used for final smoothing of sharp edges or bone spicules before the completion of surgery
    - Teeth of file works well only in **pulling stroke**

Instruments to Remove Soft Tissue From Bony Cavities

- ***<u>Curette</u>***

Angled double ended instrument to remove granulomas, small cysts, and debris from bony defect

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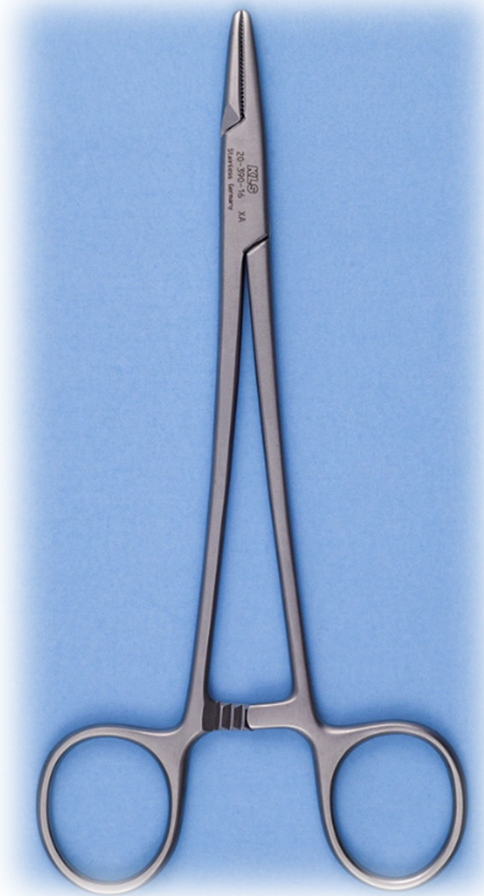
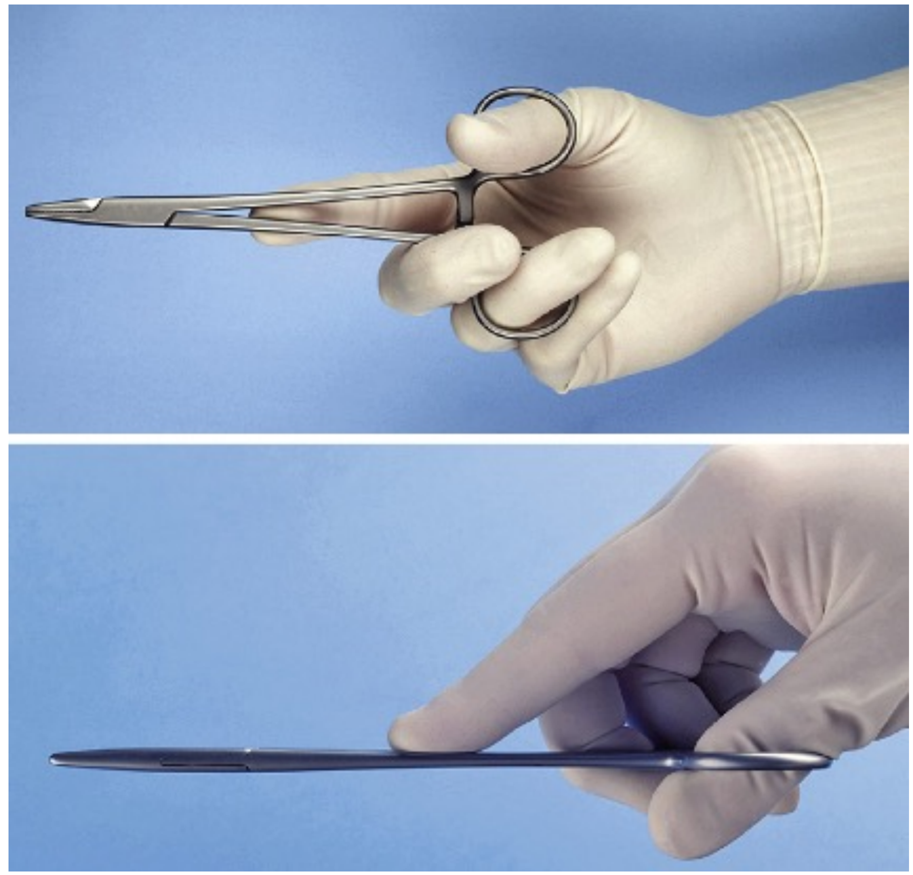
Instruments to Suture Soft Tissue

- ***<u>Needle Holder</u>***
  - has a locking handle and a short, blunt beak
  - face of the beak is cross-hatched to ensure a positive grip on the needle



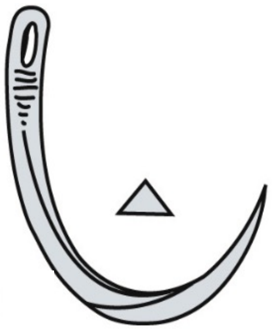
Instruments to Suture Soft Tissue

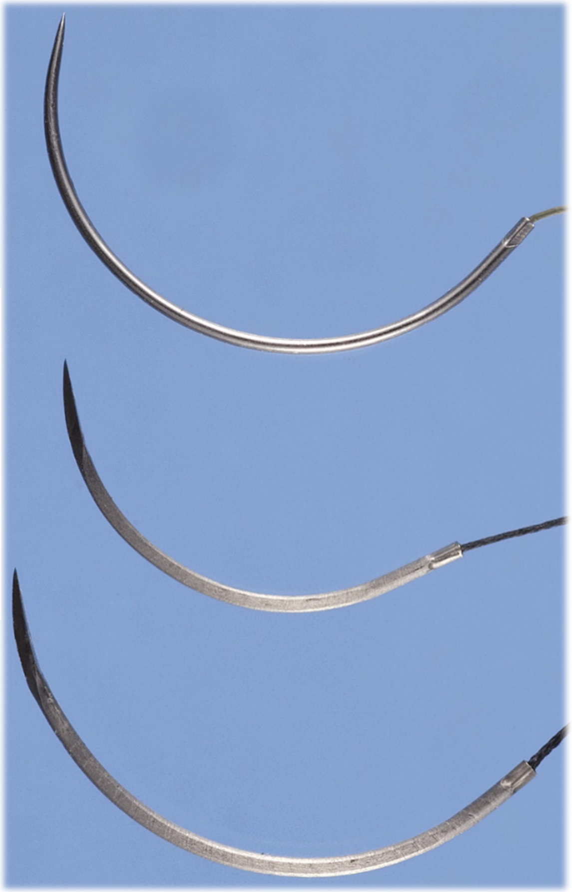
- ***<u>Handling Needle Holder</u>***
  - is held by thumb and ring finger in rings.
  - first finger is placed on top of the instrument and second finger is resting on the side of the instrument to allow better control



Instruments to Suture Soft Tissue

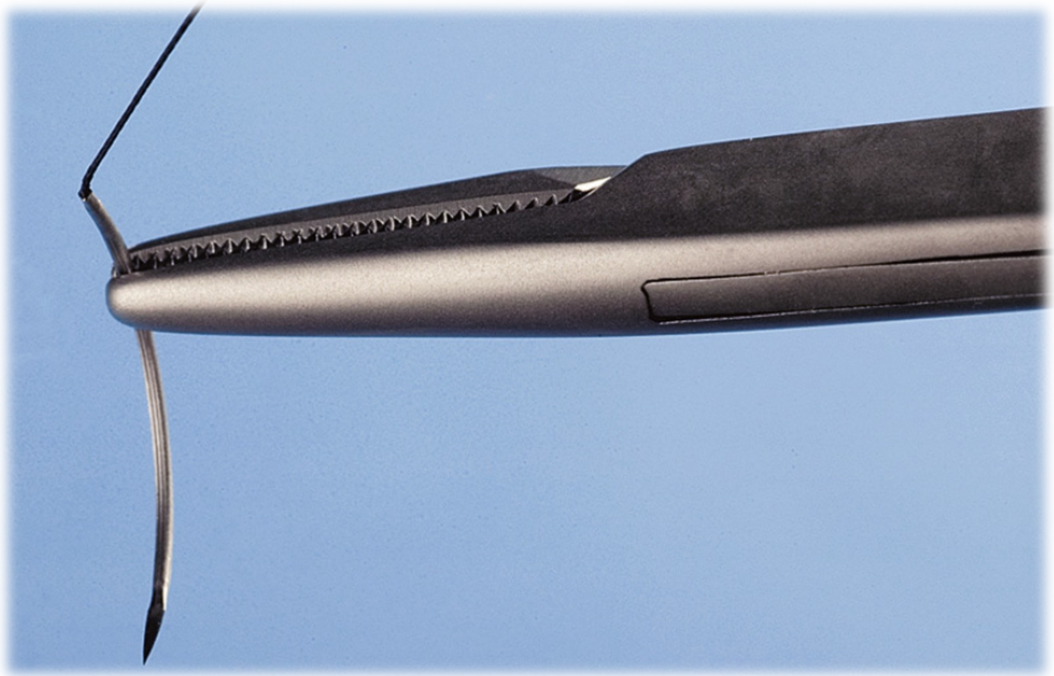
- ***Suture Needles (in OS)***
  - half-circle or three-eighths–circle
  - variety of shapes from very small to very large
  - tips of suture needles are either
    - tapered (like sewing needles)
    - triangular tips (cutting needle)
  - cutting portion of the needle extends one-third the length of the needle





Instruments to Suture Soft Tissue

- ***Suture Needle***
  - is held approximately two-thirds of the distance from the tip of the needle



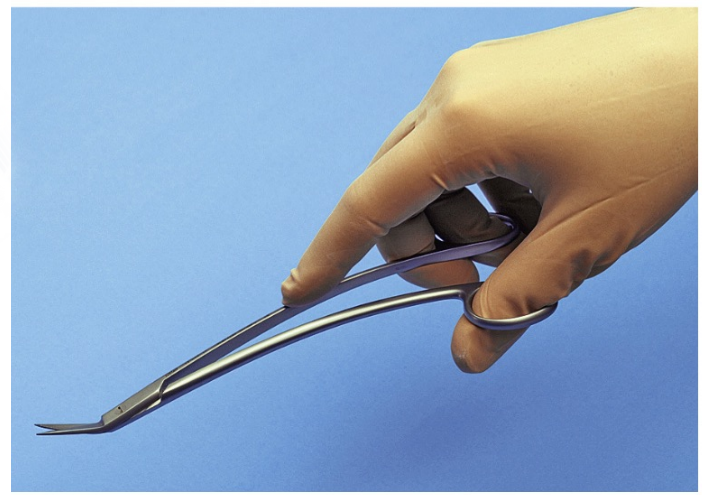
Instruments to Suture Soft Tissue

- *__Suture Materials__*
  - Classified
    - By diameter [3-0*/ (000)]
    - Resorbability
      - Resorbable
        - Gut/catgut/chromic gut*
        - Polyglycolic/polylactic acid
      - Non-resorbable (silk*, nylon, vinyl, Teflon, etc.)
    - Filament type
      - Monofilament
      - Polyfilament

Instruments to Suture Soft Tissue

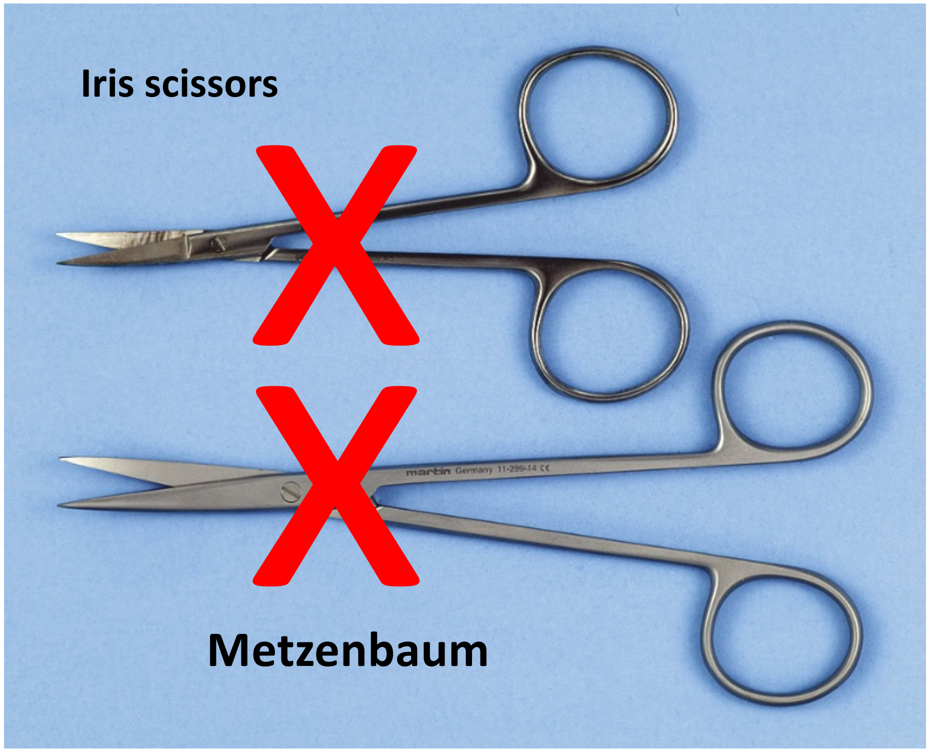
- ***Suture Scissors***
  - Have short cutting tips
  - Are held in the same fashion as the needle holder

Spencer Scissors



Instruments to Suture Soft Tissue

- ***<u>Soft Tissue Scissors</u>***
  - Iris
  - Metzenbaum



Instruments to Hold the Mouth Open

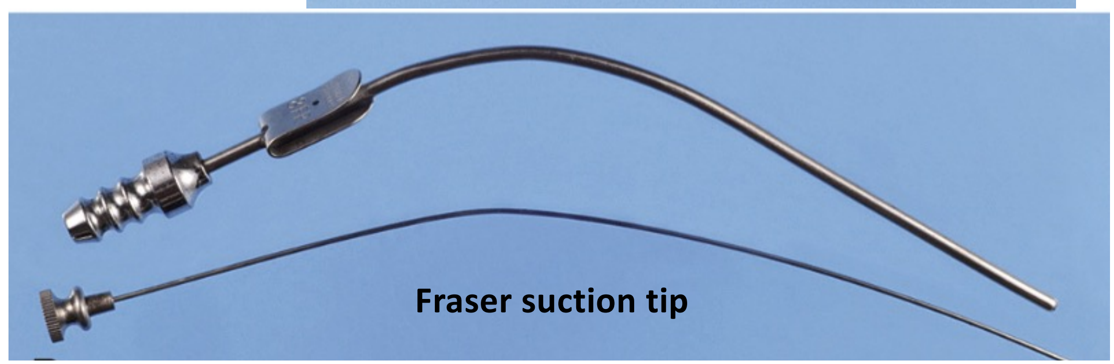
- ***Bite blocks***

Instruments to Hold the Mouth Open

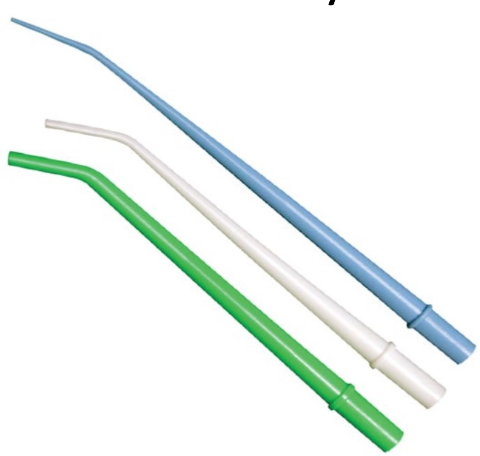
- ***<u>Molt/Side Action Mouth Prop</u>***
  - May be used
    - In sedated patients
    - In cases of mild trismus

Instruments to Remove Fluids

- *<u>Surgical Suction Tips</u>*
  - Remove blood, saliva, irrigation solutions
  - Have smaller orifice to remove fluids efficiently



Fraser suction tip



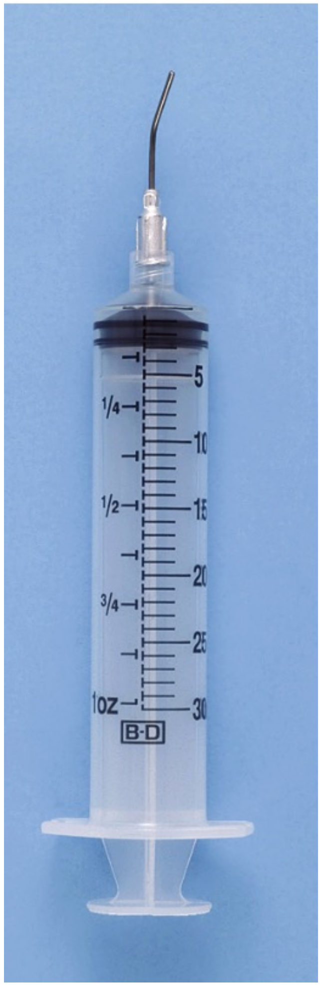
Disposable Surgical Aspirating Tips

Instruments to Hold Towels and Drapes in Position

- *<u>Towel Clip</u>*
  - ends of the towel clip can be sharp or blunt

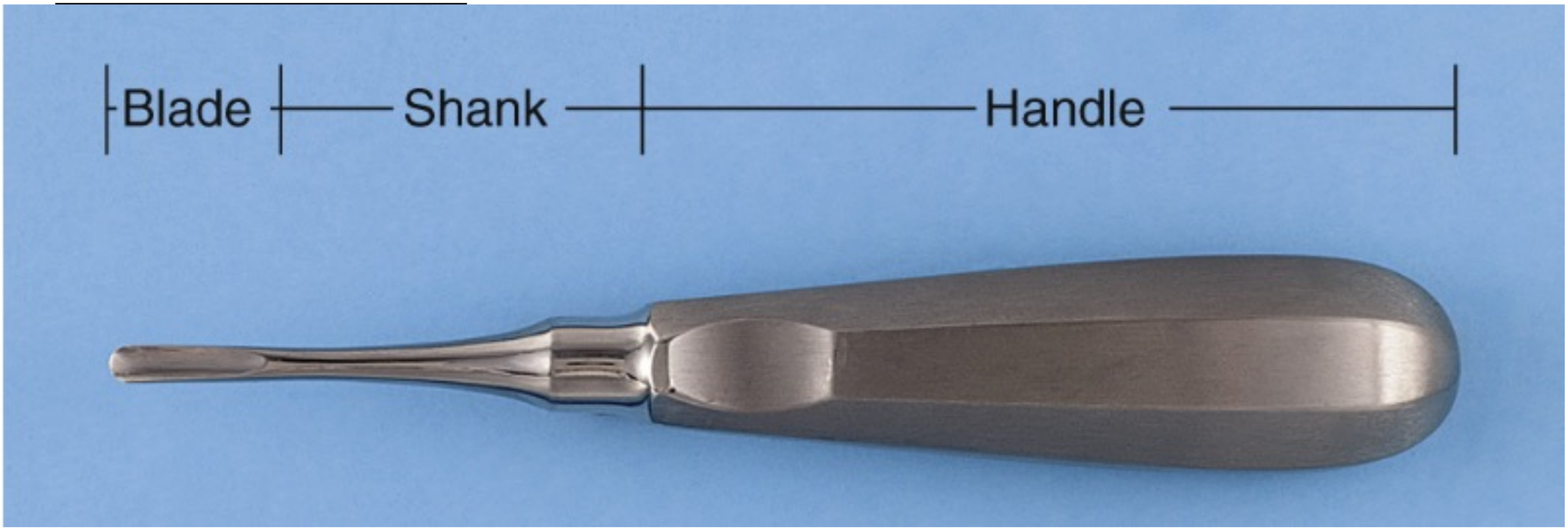
Irrigating

- ***<u>Irrigation Syringes</u>***
  - Removes debris from surgical site
  - Removes debris from bur flutes
  - Prevents heat build up and bone damage
  - Saline or Sterile Water are used as irrigating solutions



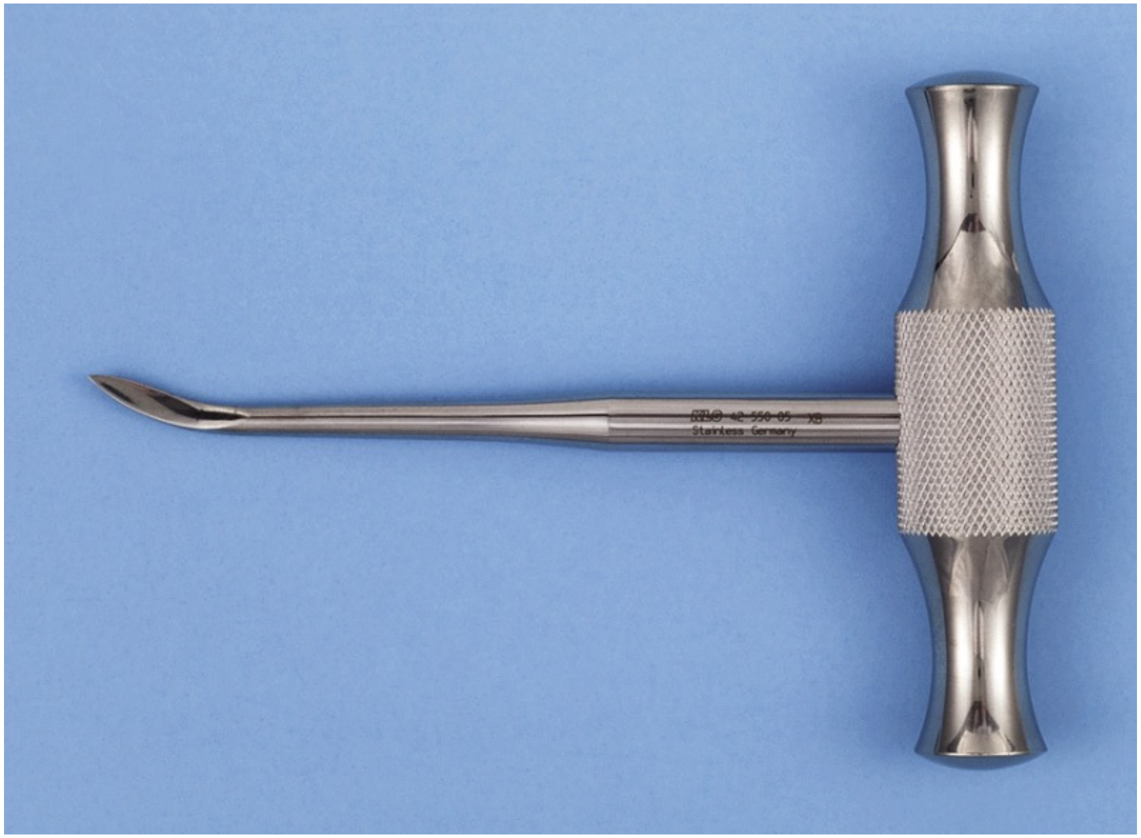
Instruments to Extract Teeth

- ***Dental Elevators***



Instruments to Extract Teeth

- ***Dental Elevator***
    - Crossbar handle can generate large amounts of force and therefore must be used with great caution



Instruments to Extract Teeth

Types of Elevators

- *Straight Elevators*
  - Most frequently utilized
  - Blade is concave on the working side

Instruments to Extract Teeth

Types of Elevators

- *Angled Elevators*
  - Most frequently utilized in posterior of the mouth
  - Miller (top image) and Potts (lower images)

Instruments to Extract Teeth

Types of Elevators

- ***<u>Triangular Elevators</u>***

  (CRYER)

  - Comes in pairs: Right and left or “East and West”

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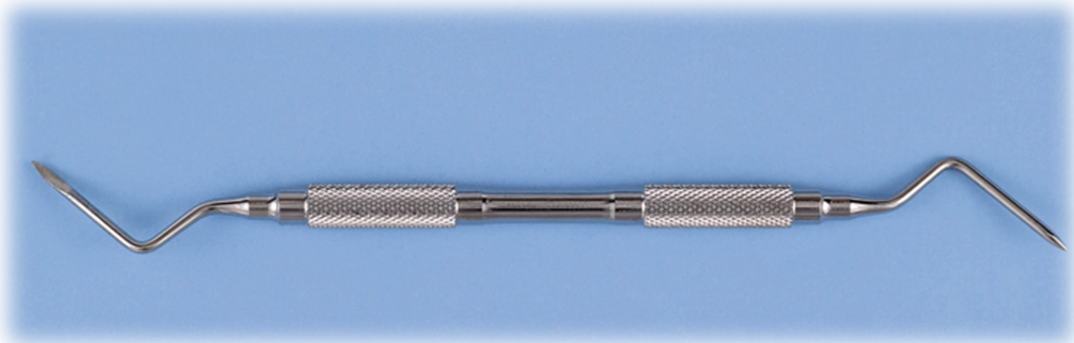
Instruments to Extract Teeth

Types of Elevators

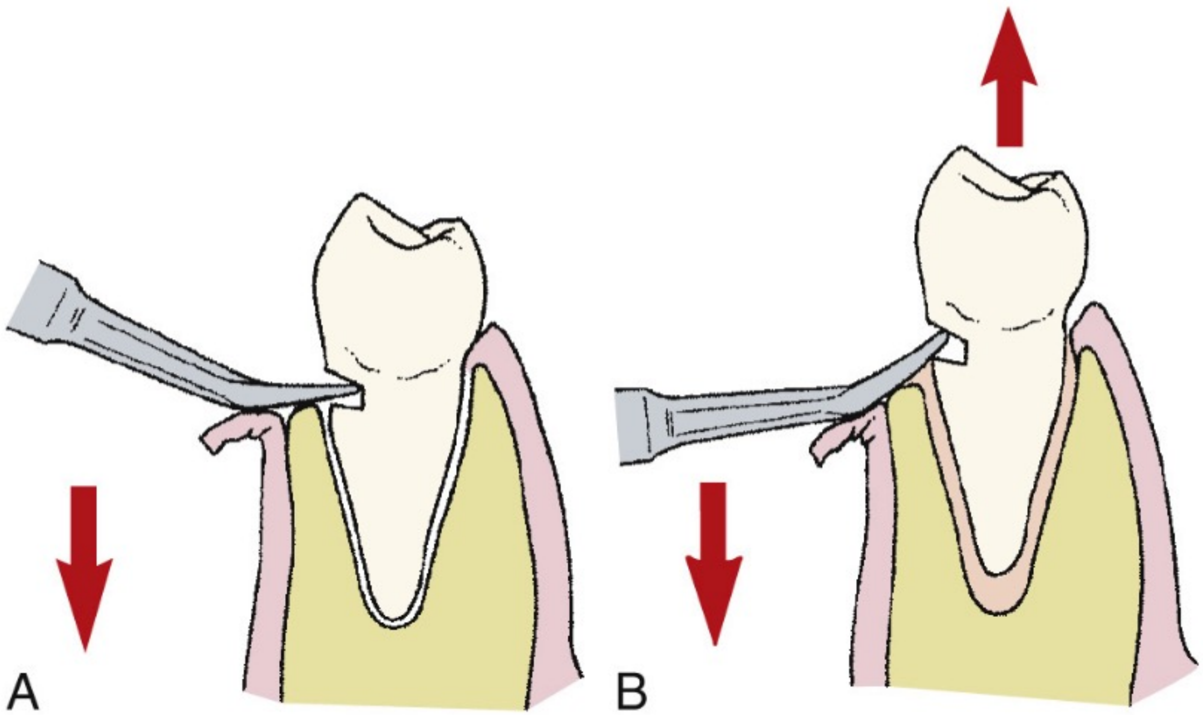
- ***Pick-Type Elevators***
  - Crane pick (top)
  - Root-tip pick or the apex elevator
    - teases small root tips from sockets



Instruments to Extract Teeth

Types of Elevators

- ***Crane pick***

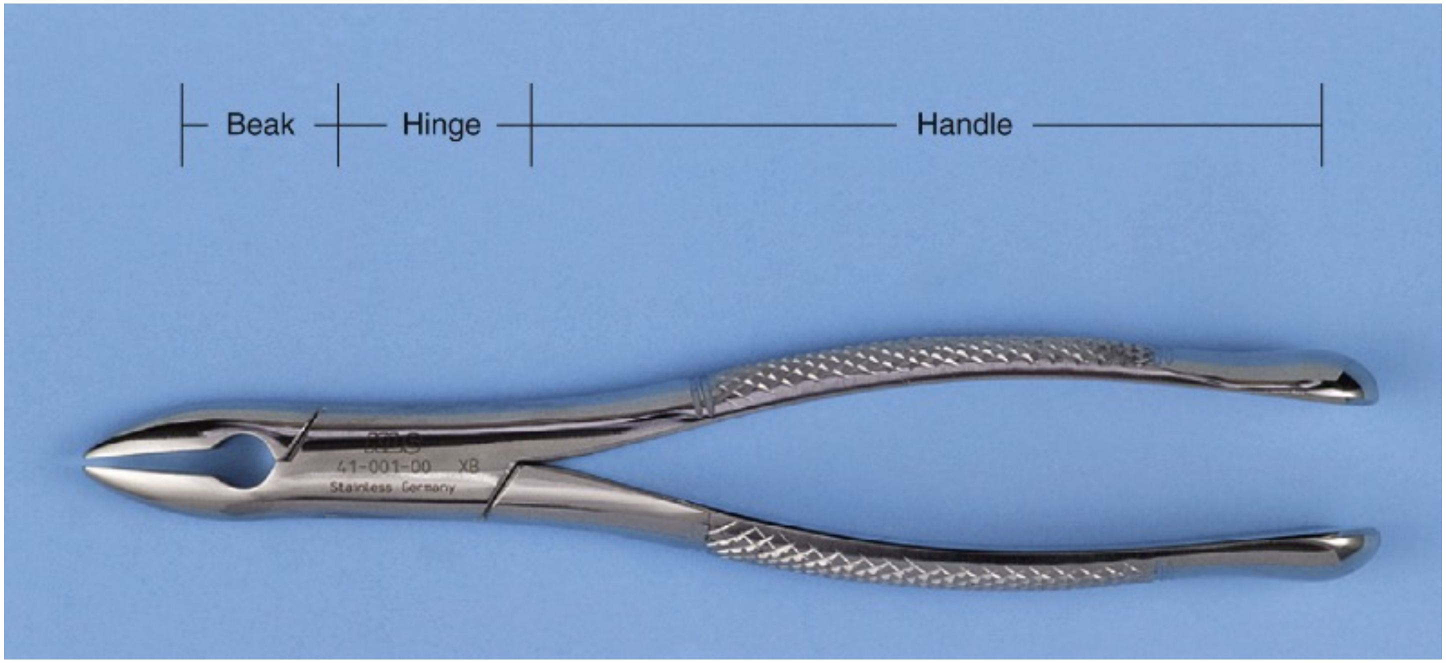


Instruments to Extract Teeth

- *Periotomes*
  - Used for ”atraumatic” extraction
  - Preserves socket anatomy
  - Severs periodontal ligaments

Instruments to Extract Teeth

- Extraction Forceps



Instruments to Extract Teeth

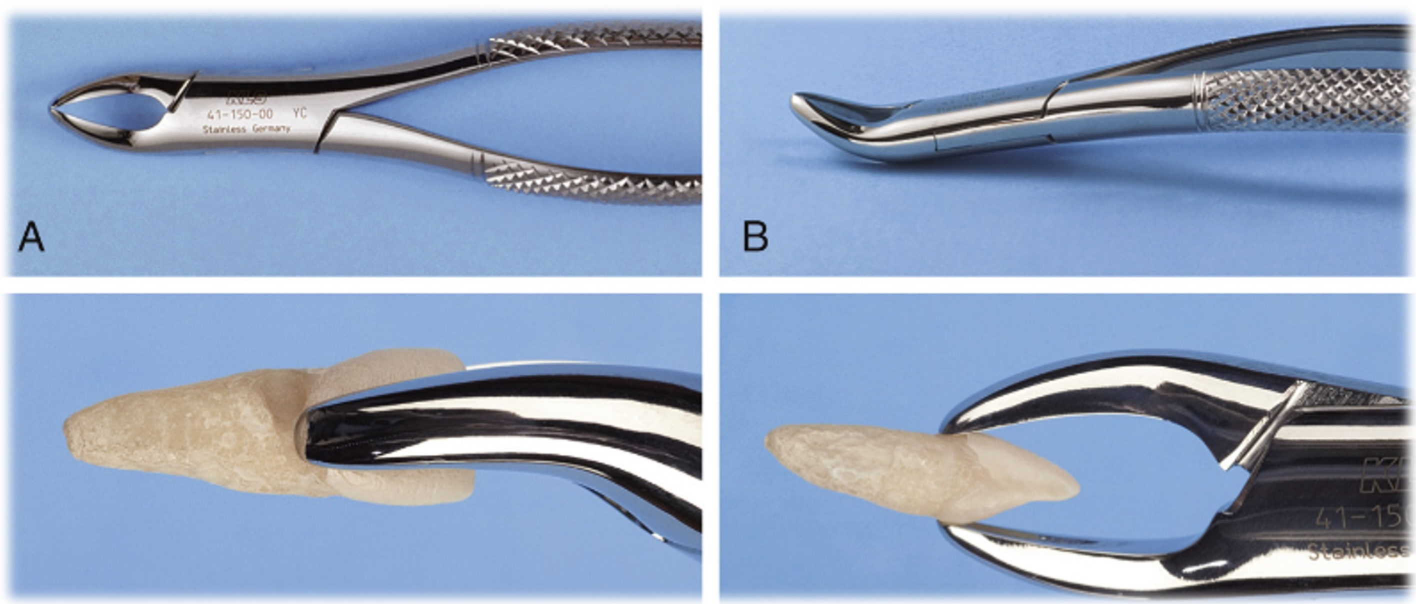
- Extraction Forceps

- *Curved handles*

Instruments to Extract Teeth

Extraction Forceps (Maxillary)

- ***<u>Upper Universal Forceps (No. 150)</u>***
  - Most frequently used to remove single-rooted max. teeth



Instruments to Extract Teeth

Extraction Forceps (Maxillary)

- ***<u>Upper Primary Teeth Universal Forceps (No. 150S)</u>***
  - smaller version of 150 which adapt well to all maxillary primary teeth

Instruments to Extract Teeth

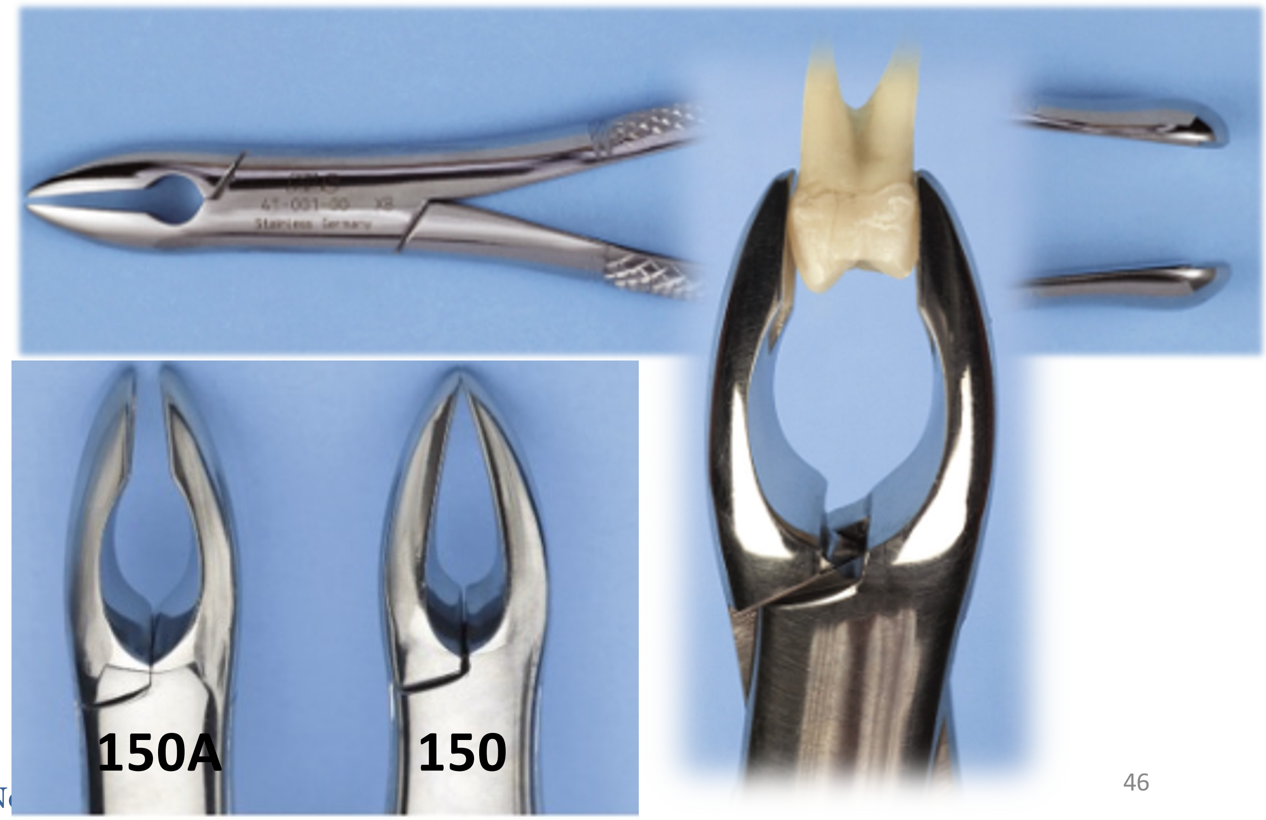
Extraction Forceps (Maxillary)

- *No 1 Forceps*
  - Fits roots of maxillary incisors and canines

Instruments to Extract Teeth

Extraction Forceps (Maxillary)

- *__No. 150A forceps__*
  - Beak adapt better to maxillary premolar
  - Poor adaptation to anterior teeth



Instruments to Extract Teeth

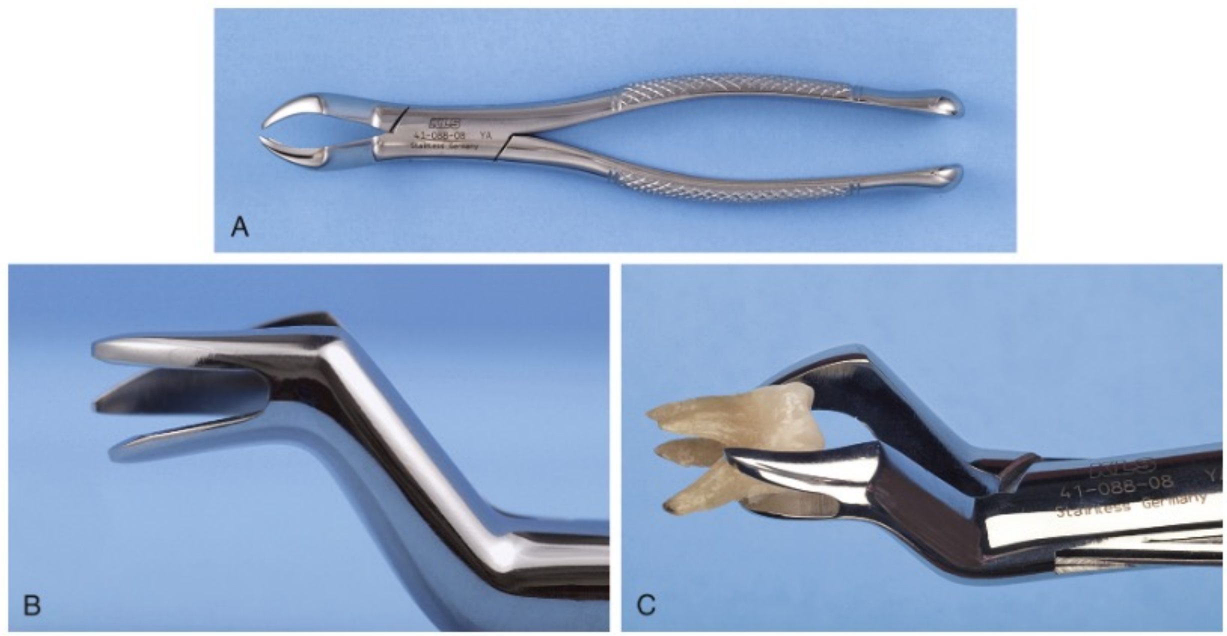
Extraction Forceps (Maxillary)

- ***No. 53R and 53L***
  - Fit the upper molars
  - One side fits into buccal furcation
  - Concave side fits palatal root

Instruments to Extract Teeth

Extraction Forceps (Maxillary)

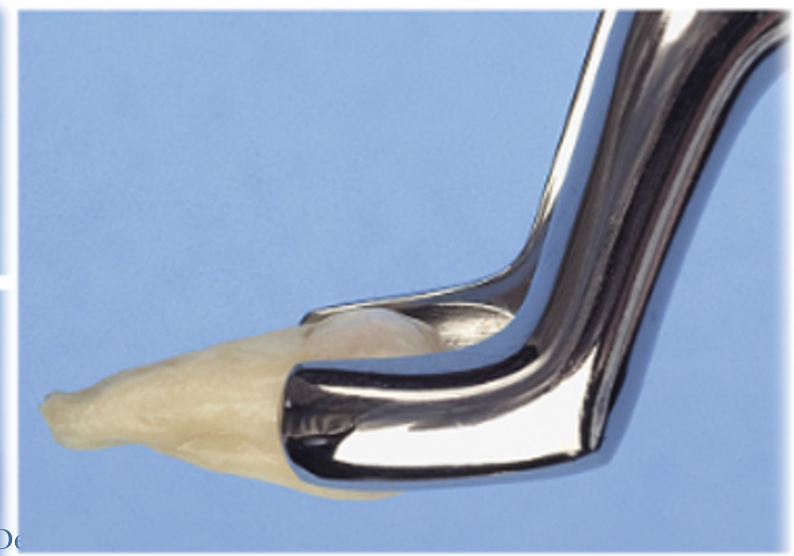
- *No. 88R and 88L*
  - Fit the upper molars
  - Accentuated pointed beak
  - Useful when crown of upper molar is severely carious



Instruments to Extract Teeth

Extraction Forceps (Maxillary)

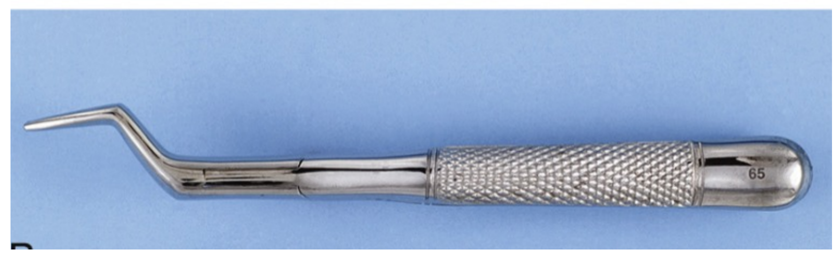
- ***<u>No. 210S forceps</u>***
  - Used for extraction of second molars and erupted third molars that have a single conical root

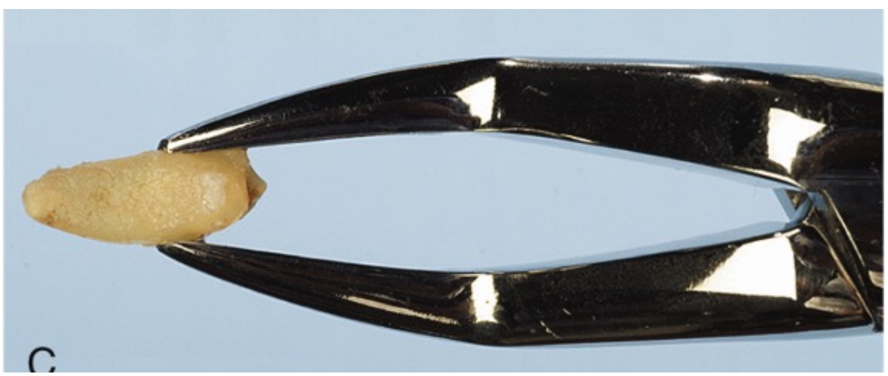


Instruments to Extract Teeth

Extraction Forceps (Maxillary)

- ***<u>No. 65 forceps</u>***
  - Used to remove broken maxillary molar roots
  - Can be used with ant. Roots too





Instruments to Extract Teeth

Extraction Forceps (Mandibular)

- ***<u>No. 151 (Universal Lower Forceps)</u>***
  - Used to remove single-rooted lower teeth
  - Resemble 150, except the beaks point downward

Instruments to Extract Teeth

Extraction Forceps (Mandibular)

- ***No. 151S (Universal Lower Forceps for primary teeth)***
  - Resemble 151, scaled down to adapt to lower primary teeth

Instruments to Extract Teeth

Extraction Forceps (Mandibular)

- ***<u>No. 151A</u>***
  - Beaks are parallel for better adaptation to mandibular premolars

Instruments to Extract Teeth

Extraction Forceps (Mandibular)

- ***<u>English style of vertical-hinge forceps</u>***
  - Used for single-rooted lower teeth
  - While using, great care must be exercised to prevent root fracture


Instruments to Extract Teeth

Extraction Forceps (Mandibular)

- ***No. 17 molar forceps***
- Beaks enter the bifurcation of lower molars
- Adapts well to right and left molars

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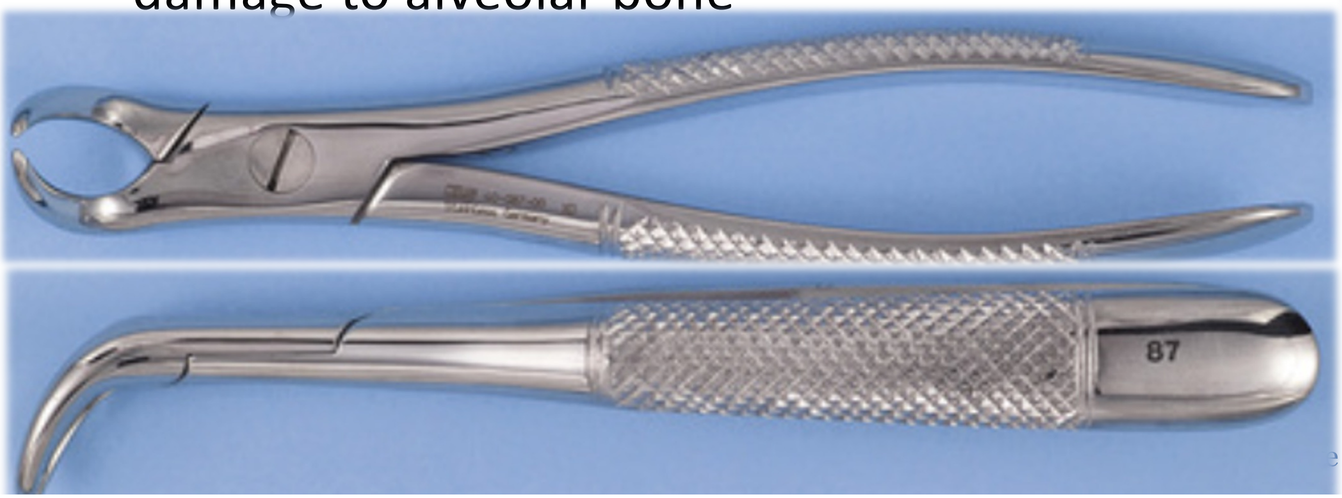
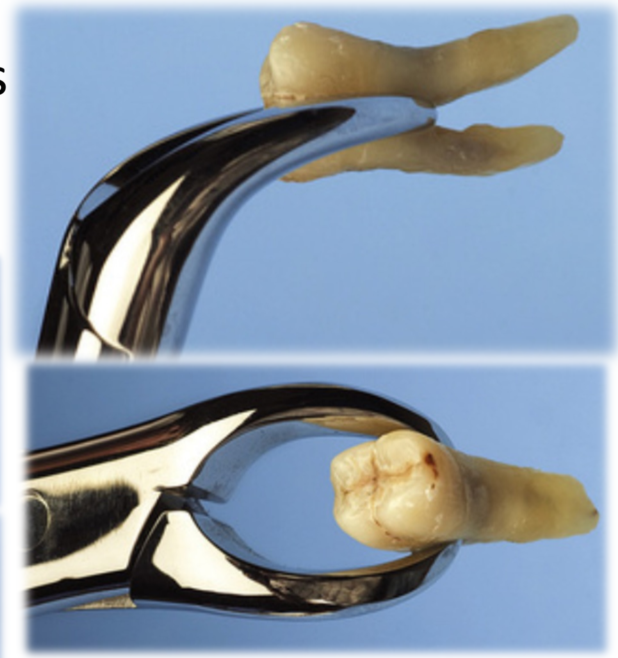
Instruments to Extract Teeth

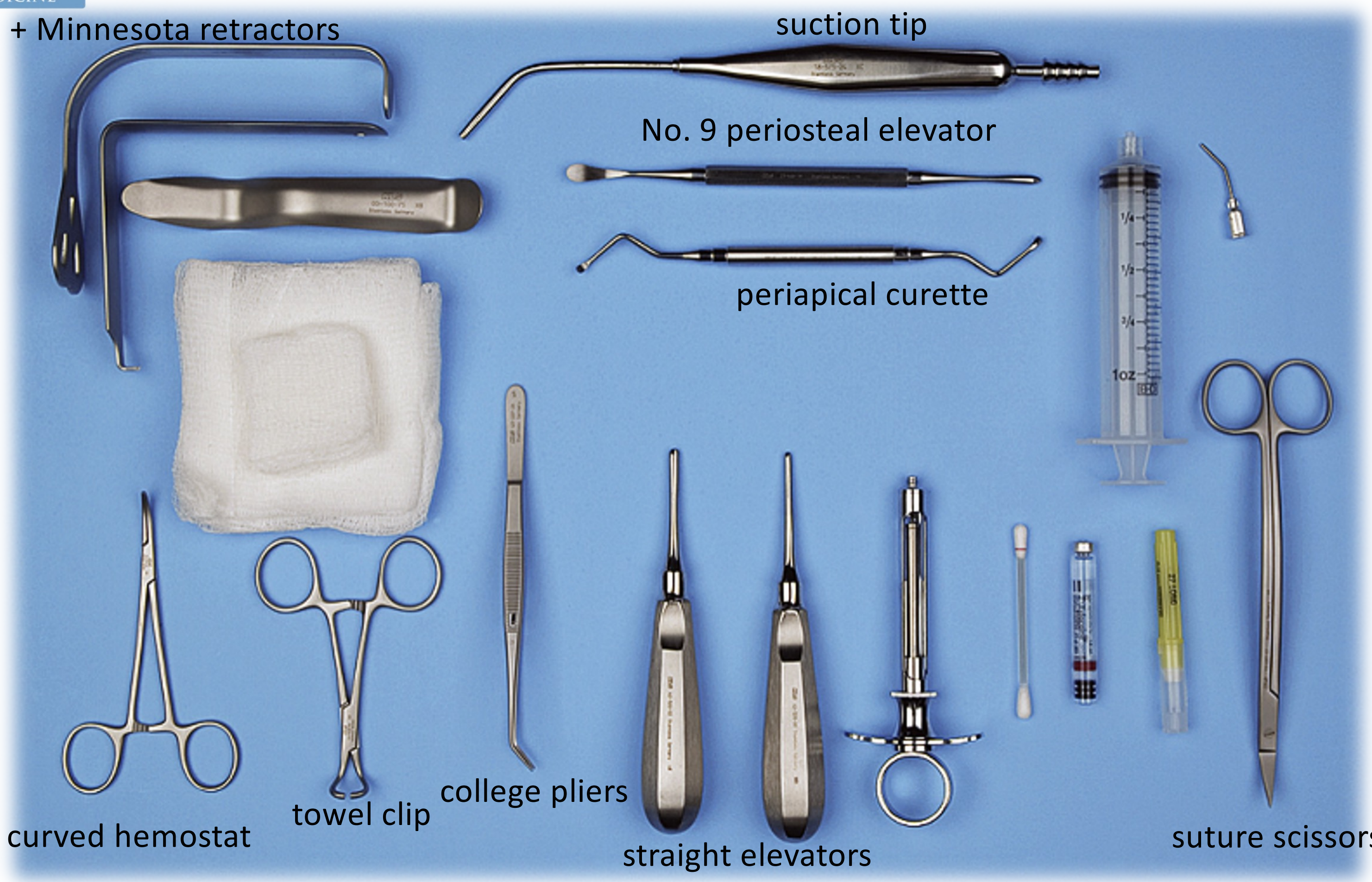
Extraction Forceps (Mandibular)

- ***<u>No. 87 forceps (cowhorn forceps)</u>***
  - Heavy beaks enter the bifurcation of lower molars
  - Use with caution to prevent root fracture and damage to alveolar bone





Basic Extraction Tray



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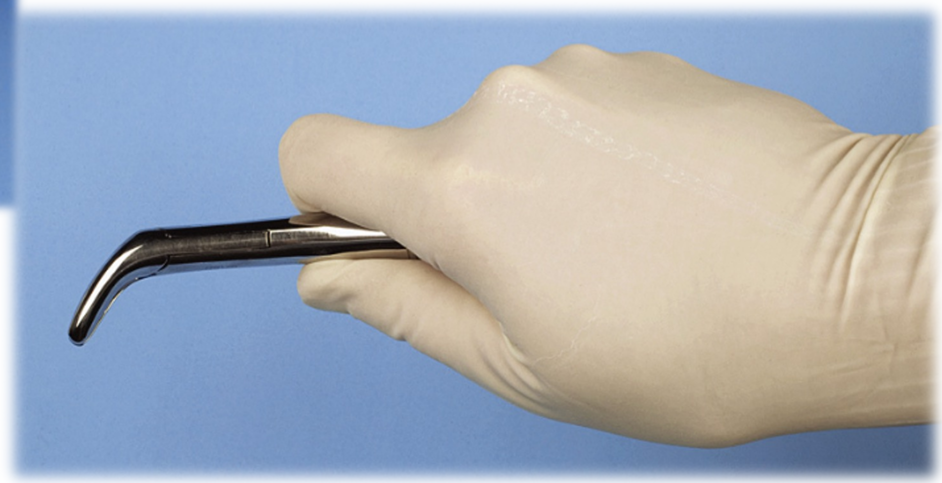
Surgical Extraction Tray

Biopsy Tray

Postoperative Tray

Holding Maxillary Forceps

Holding Mandibular Forceps



Holding English Style Forceps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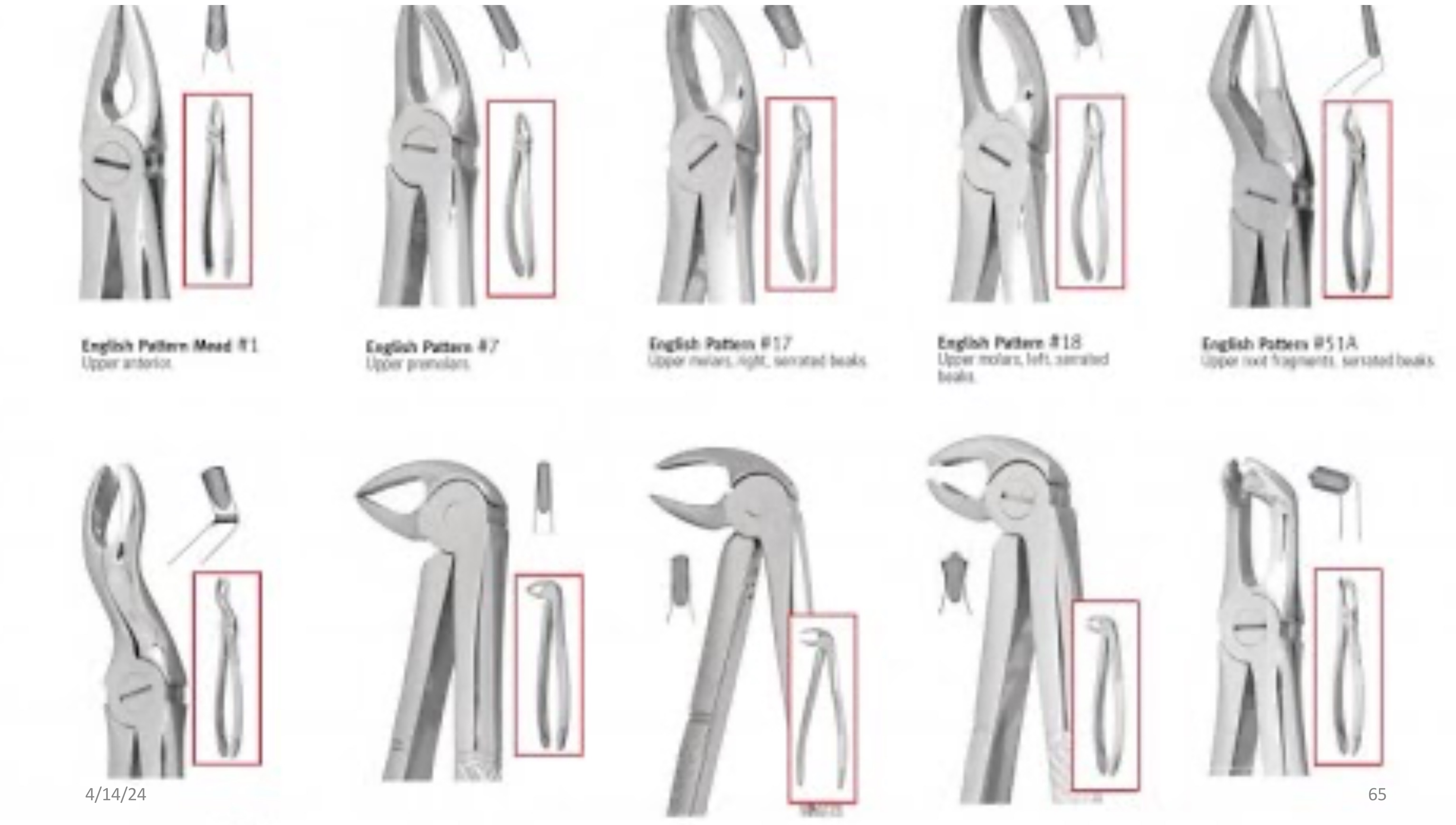
English Pattern Mead #1
Upper anterior.
English Pattern #7
Upper premolars.
English Pattern #17
Upper molars, right, serrated beaks.
English Pattern #18
Upper molars, left, serrated beaks.
English Pattern #51A
Upper root fragments, serrated beaks.


Best Positioning
for Exodontia

Thank You for Your Attention!